



Introduction to Unix

Biodiversity Genomics
Mendoza, Argentina, 2025

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Crash course

- Introduction
 - Why command line?
 - Terminology
 - GUI and GLI
 - Unix
 - The Shell (bash, zsh)
 - The terminal
 - The UNIX filesystem
 - Running programs
 - Getting help
 - Best practices
- Practical



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Why command line?

- Automation
 - Easy to work with multiple files (10 000s)
 - In parallel
- Efficient handling of large files
 - Exploring large files with simple commands
 - Genomics data are huge GB, TB
- High-performance computing (HPC) is the use of supercomputers and computer clusters, requires command line interaction
- Reproducibility
 - Repeat the same analyses exactly the same way across samples and across studies

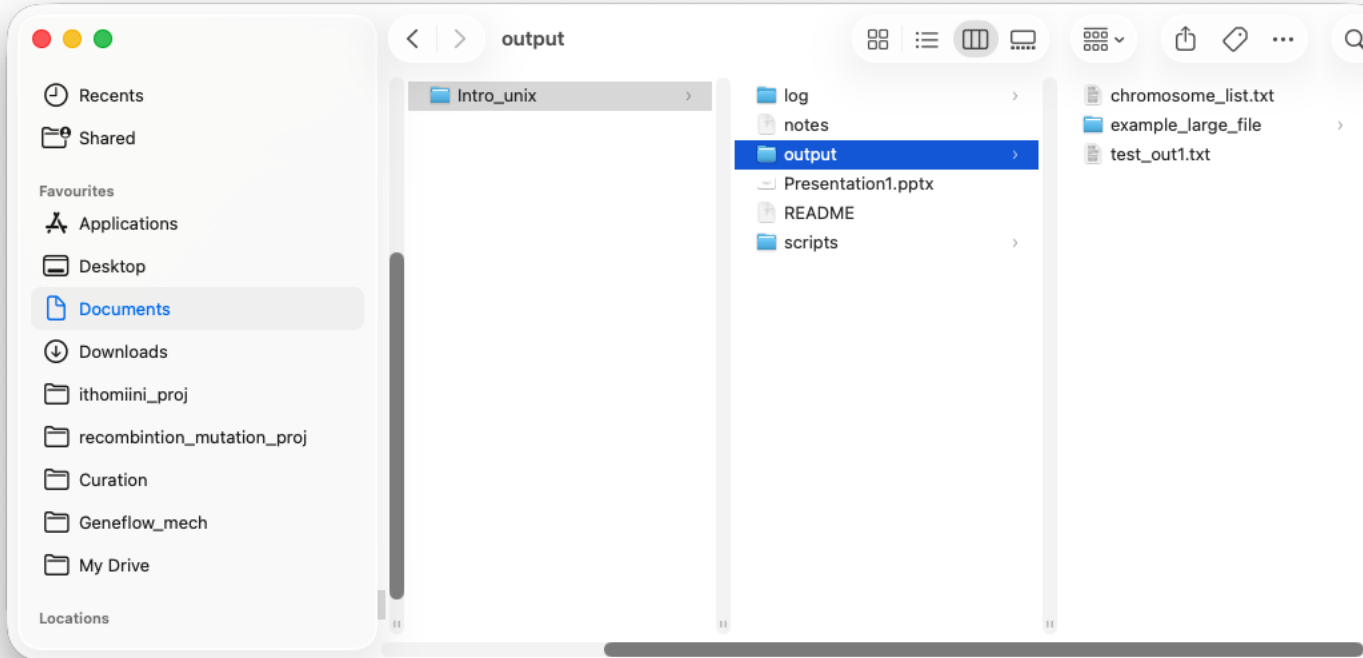
Terminology

- GUI and CLI
- UNIX
- Shell
- The terminal

Difference between GUI and CLI.

- Graphical User Interface (GUI)
 - Point and click (almost)
- Command Line Interface (CLI)
 - You have to type the commands

GUI (Finder)



CLI (find)

```
Intro_unix -- zsh -- 72x23
(base) kn9@mib118791s Intro_unix % find .
.
./.DS_Store
./Presentation1.pptx
./output
./output/example_large_file
./output/example_large_file/ilMelLudo1.1.primary.fa.gz
./output/test_out1.txt
./output/chromosome_list.txt
./README
./notes
./scripts
./scripts/not_so_useful_scripts
./scripts/.DS_Store
./scripts/useful_scripts
./scripts/useful_scripts/get_fasta_headers.sh
./log
./~$Presentation1.pptx
(base) kn9@mib118791s Intro_unix %
```

What is UNIX?

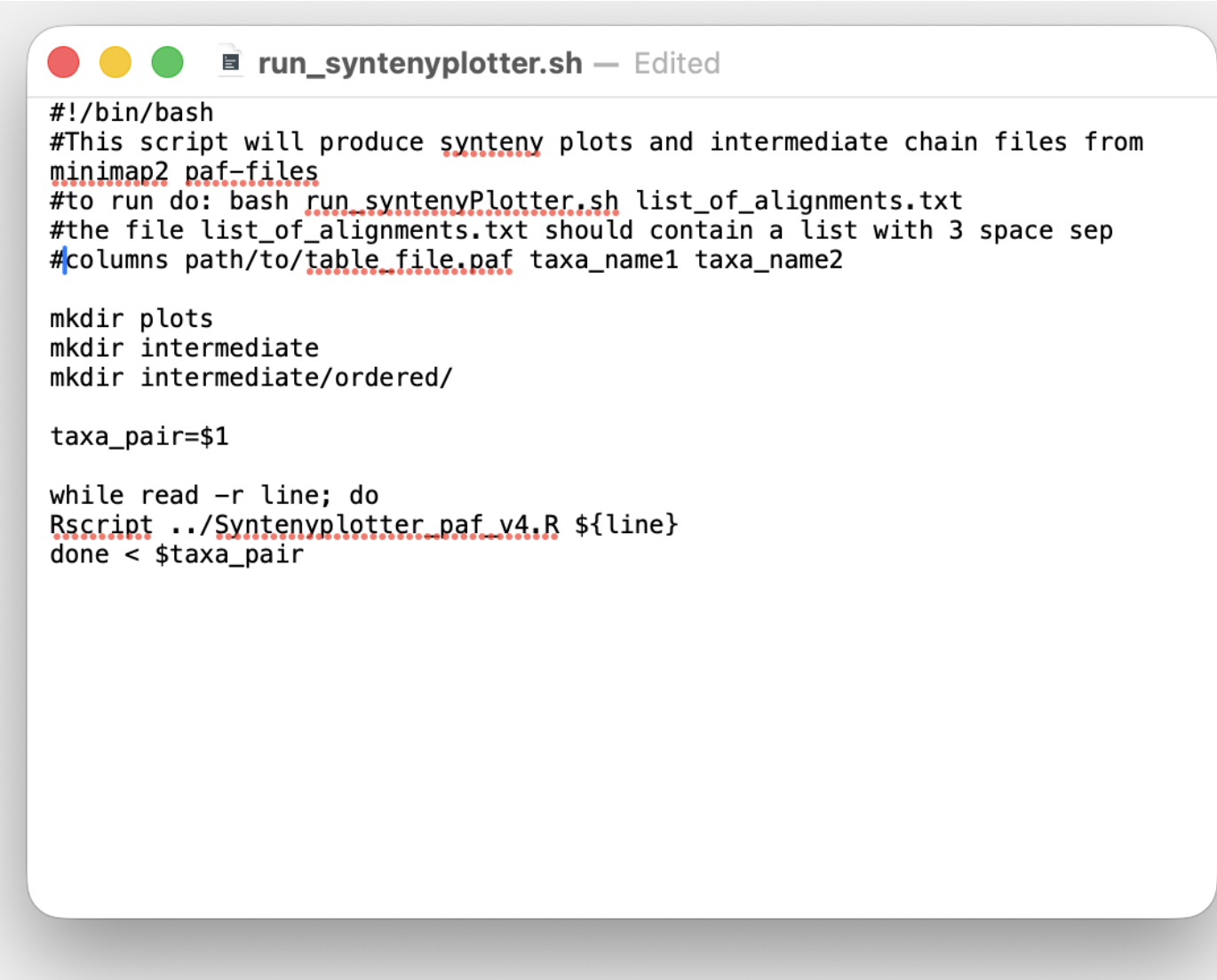
- UNIX is an operating system (OS)
 - the OS controls the interaction, activity and resource distribution in a computer
- UNIX was developed in the 1970s
- Modular design – multiple small programs that can be combined to perform complex pipelines
- UNIX way of interaction with the computer on the command line and its hierarchical file system became a standard
- Today there are many Unix-like operating systems (MacOS, Linux)
 - Different applications that allows Unix style interaction with Windows



The AT&T UNIX-PC, released in 1985. (Photo: OldComputers.net)

The shell

- The shell is a language that interprets the commands we use to communicate with the computer
- Multiple variants
 - bash, zsh (open source)
- Shell script (or bash script)
 - collection of shell commands in a text file that can be executed together



```
run_syntenplotter.sh — Edited

#!/bin/bash
#This script will produce syteny plots and intermediate chain files from
minimap2 paf-files
#to run do: bash run_sytenyPlotter.sh list_of_alignments.txt
#the file list_of_alignments.txt should contain a list with 3 space sep
#columns path/to/table_file.paf taxa_name1 taxa_name2

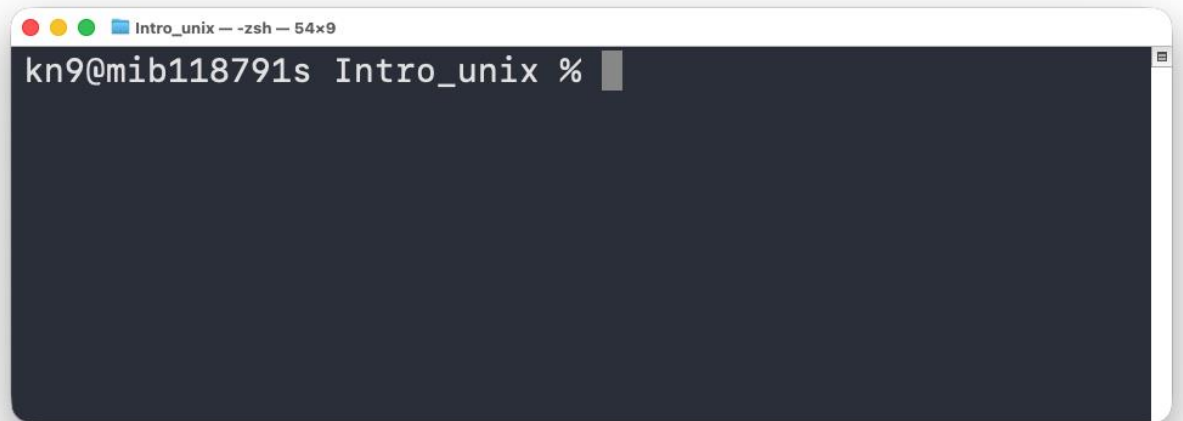
mkdir plots
mkdir intermediate
mkdir intermediate/ordered/

taxa_pair=$1

while read -r line; do
Rscript ../Sytenyplotter_paf_v4.R ${line}
done < $taxa_pair
```


The terminal environment

- The terminal is an application that gives access to a shell environment
- It is where you are typing your commands
 - Terminal (Mac)
 - MobaXterm or Command Prompt (Windows)
- The terminal can also be accessed via GUI like Visual Studio Code

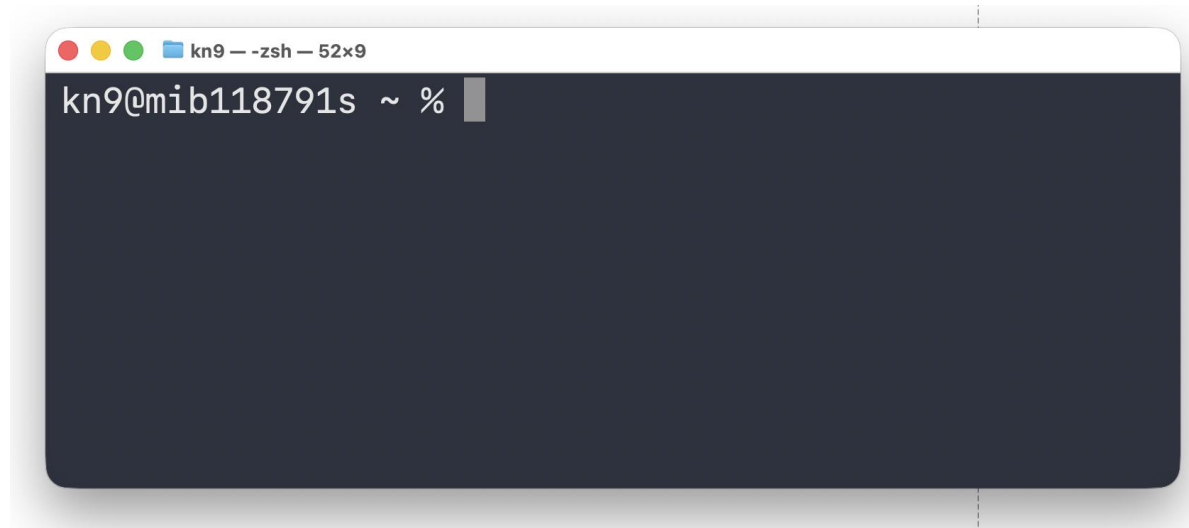


Summary

- UNIX-like system under the hood that controls the computer
- Shell is the translator between you and your computer
- The terminal is the place where you type your commands

The terminal environment - Exercise

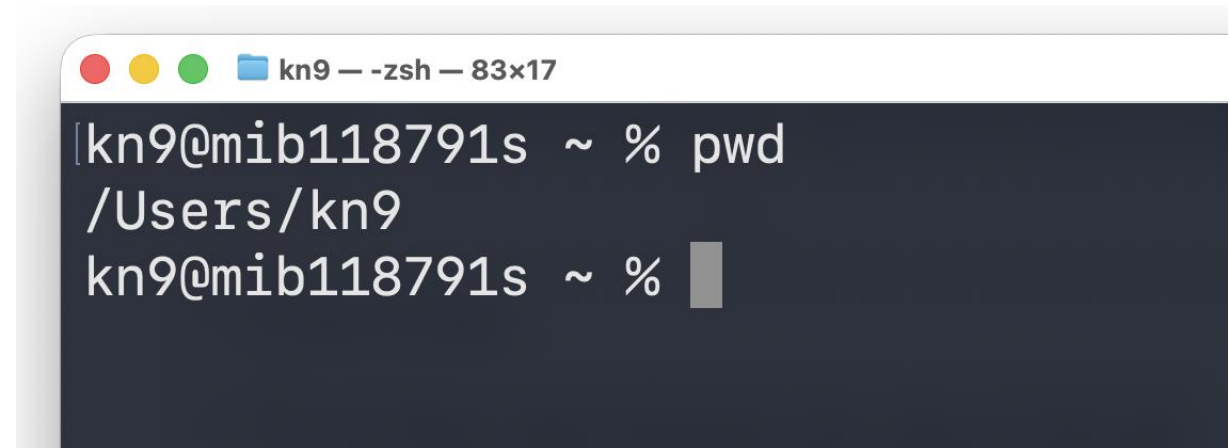
- Find the terminal on your local computer
- Check out the prompt structure:
 - username@hostname path
 - Which prompt symbol do you see?
 - \$ % >



The terminal environment

A good way to find out where you are is the command `pwd` (print working directory)

Type `pwd` in the prompt
What do you see?

A screenshot of a terminal window with a dark background and light text. The window title bar at the top shows three colored circles (red, yellow, green) followed by the text "kn9 - -zsh - 83x17". The terminal content shows the prompt "kn9@mib118791s ~ %" followed by the command "pwd" on the next line. The output of the command, "/Users/kn9", is displayed on the line following the command. A new prompt "kn9@mib118791s ~ %" is shown on the next line, followed by a grey rectangular cursor block.

```
kn9@mib118791s ~ % pwd
/Users/kn9
kn9@mib118791s ~ %
```

The terminal environment

What do you have in your local computers home directory?

The command `ls` returns a list of the content in a directory

Type `ls` in the prompt

What do you see?

Log in to server

- The terminal can also be used to connect to other computers and servers
- Connect to our server where we will be working

```
ssh genomics@toko.uncu.edu.ar
```

```
genomics@toko.uncu.edu.ar's password: ?
```

Log in to server

- The terminal can also be used to connect to other computers and servers
- Connect to our server

```
ssh genomics@toko.uncu.edu.ar
```

```
genomics@toko.uncu.edu.ar's password: genomics2025
```

Don't worry if you do not see what you are typing

The terminal environment

- You should now be on the server
- Check out the prompt structure:
 - username@hostname path
 - Is the text different from before?
- Which prompt symbol do you see?
 - \$ % >



```
kn9 — ssh -X -Y genomics@toko.uncu.edu.ar — 52x9
(base) genomics@toko:~/scratch$
(base) genomics@toko:~/scratch$ ls
backup/  data/  scripts/
(base) genomics@toko:~/scratch$
```

The terminal environment

- Log out of server

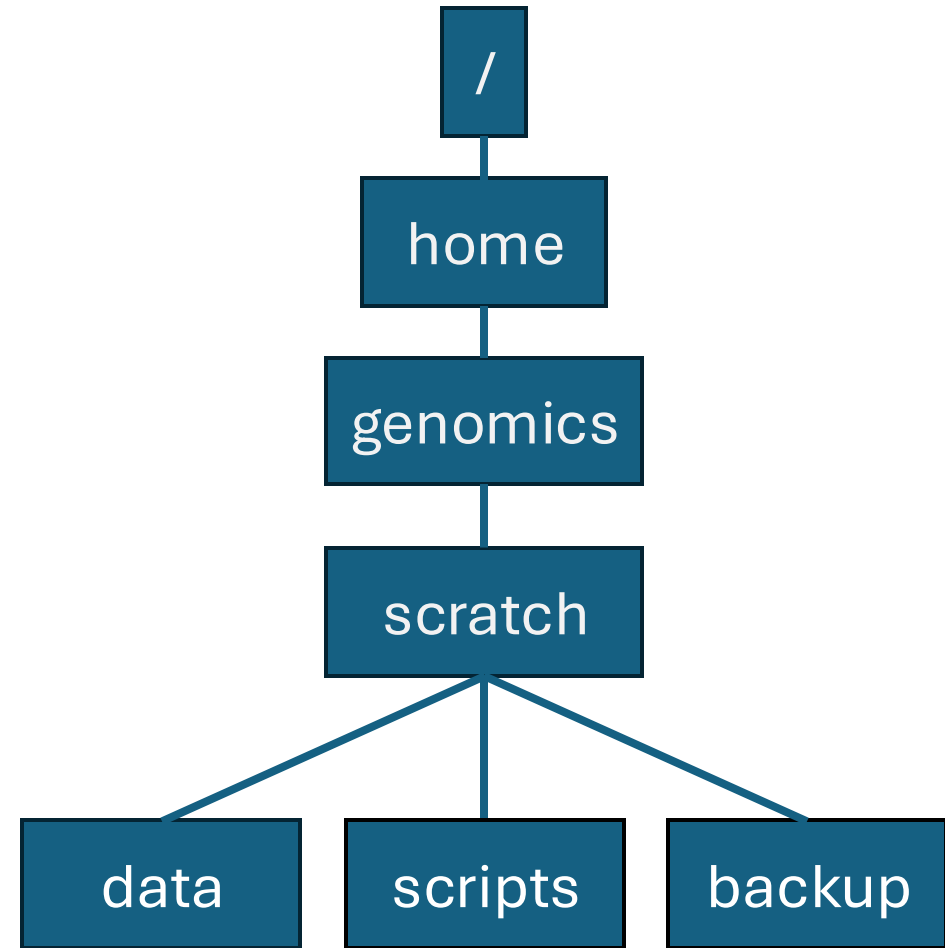
```
exit
```

- You should now be on your local computer

```
[(base) genomics@toko:~$ exit  
logout  
Connection to toko.uncu.edu.ar closed.  
kn9@mib118791s output %
```

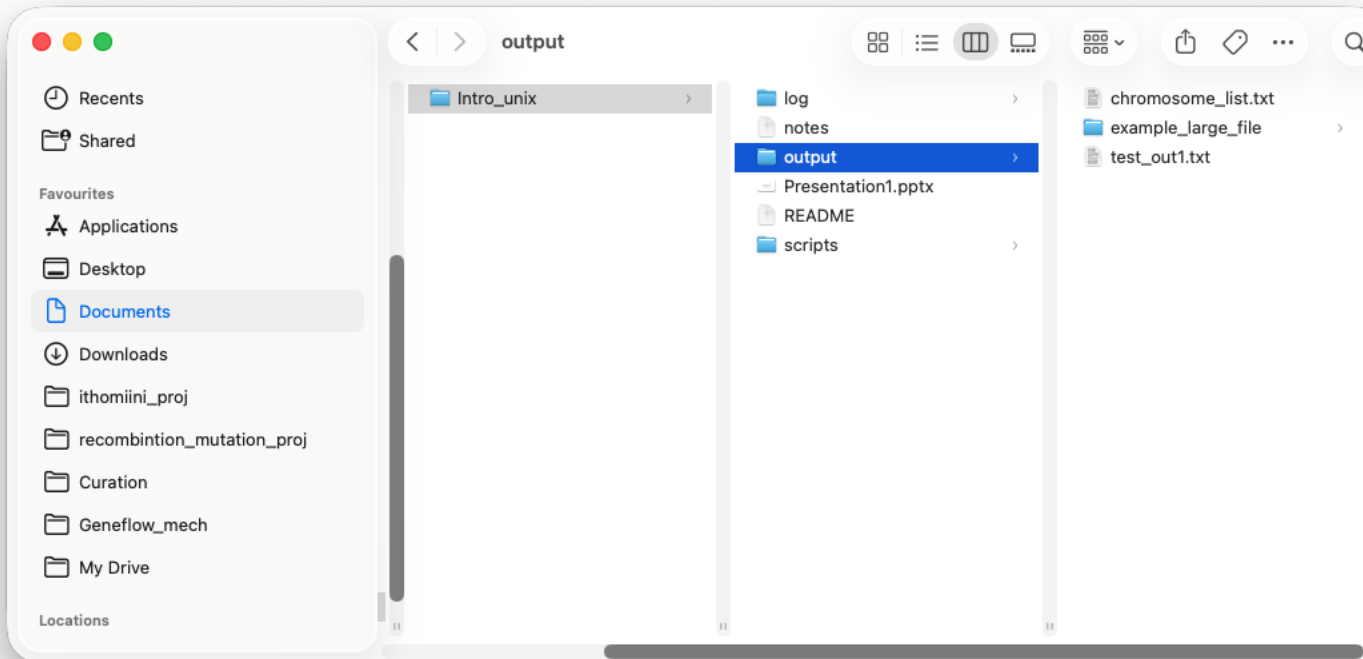
The filesystem

- Hierarchical tree structure that consists of
 - Directories $\approx$ folders
 - Directories can contain other directories and files
 - Files
 - my_text.txt, table.csv, my_script.sh)
 - (Links)



Directory structure

The filesystem



```
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[(base) kn9@mib118791s Intro_unix % find .
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./output/test_out1.txt
./output/chromosome_list.txt
./README
./notes
./scripts
./scripts/not_so_useful_scripts
./scripts/.DS_Store
./scripts/useful_scripts
./scripts/useful_scripts/get_fasta_headers.sh
./log
./~$Presentation1.pptx
(base) kn9@mib118791s Intro_unix %
```

The filesystem

- Log in to the server again

```
ssh genomics@toko.uncu.edu.ar  
genomics@toko.uncu.edu.ar's  
password: genomics2025
```

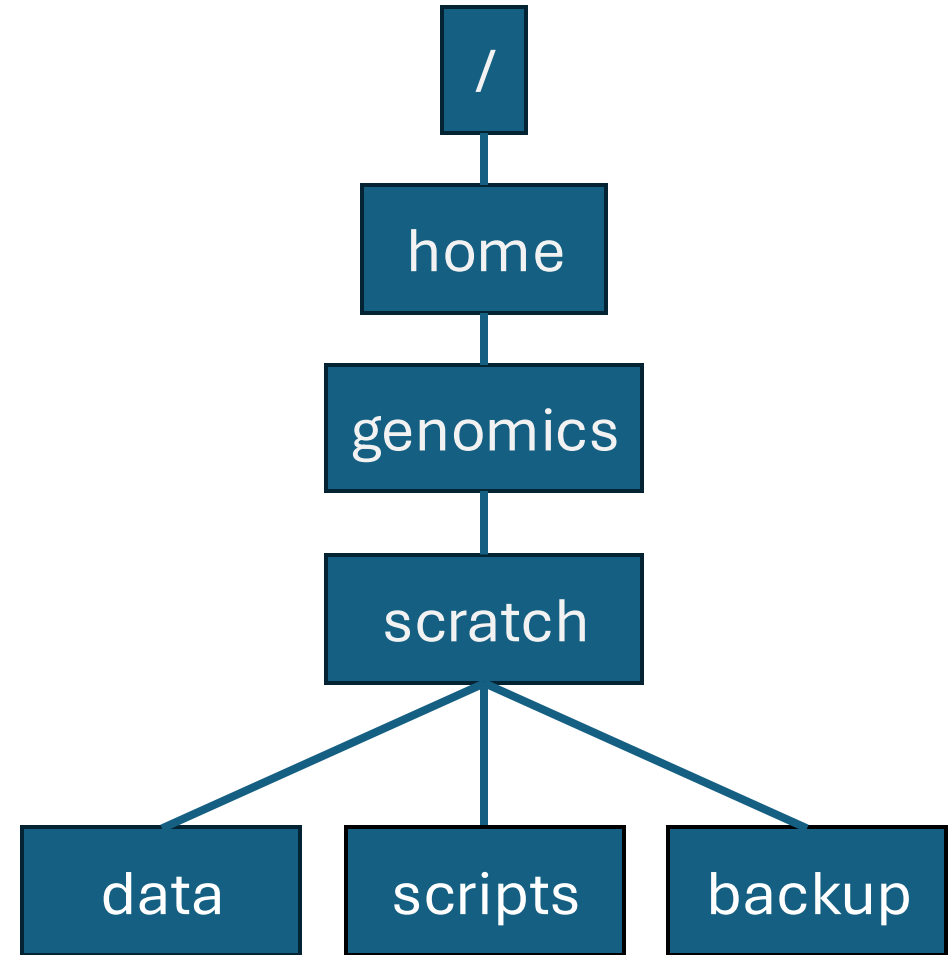
- Are you on the server?
 - Hint-check the prompt

The filesystem

- Inspect the directory structure

```
pwd
```

```
ls
```



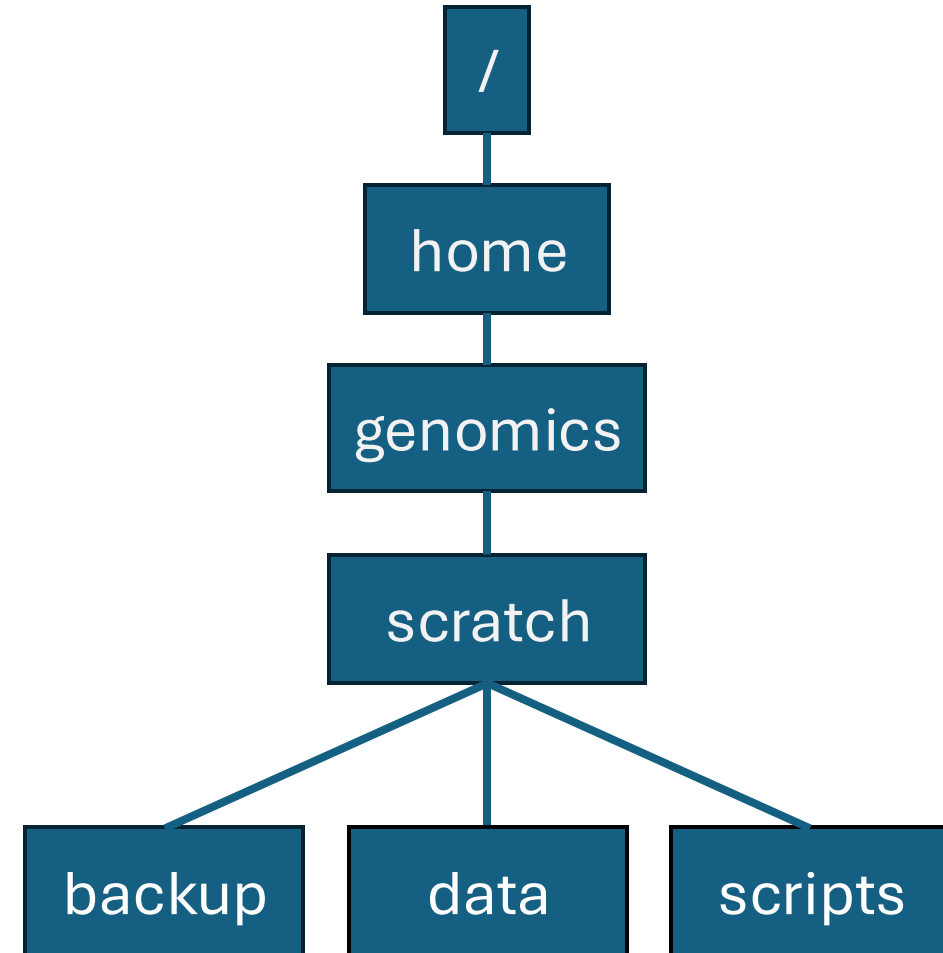
The filesystem

- Inspect the directory structure

```
pwd
```

```
ls
```

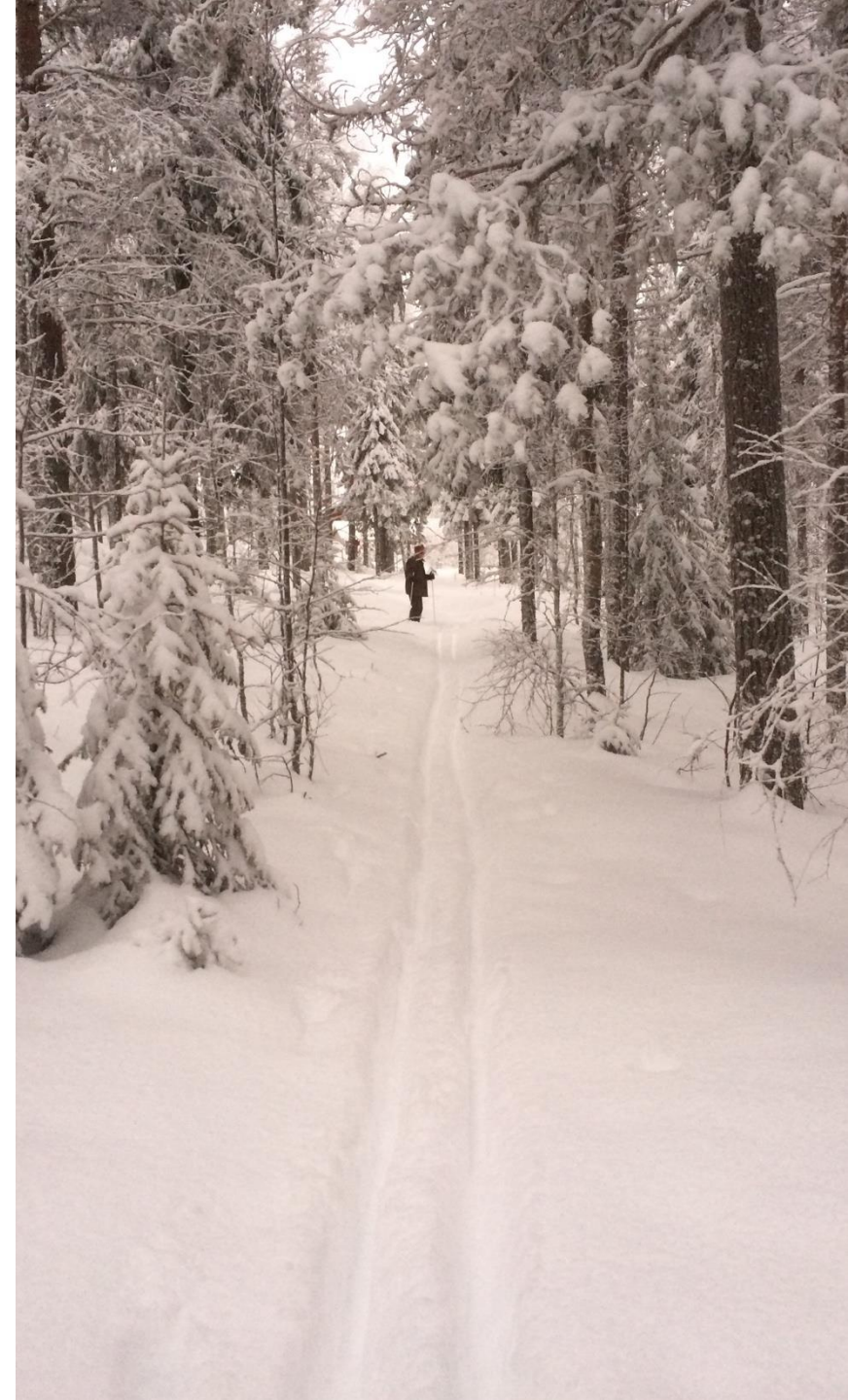
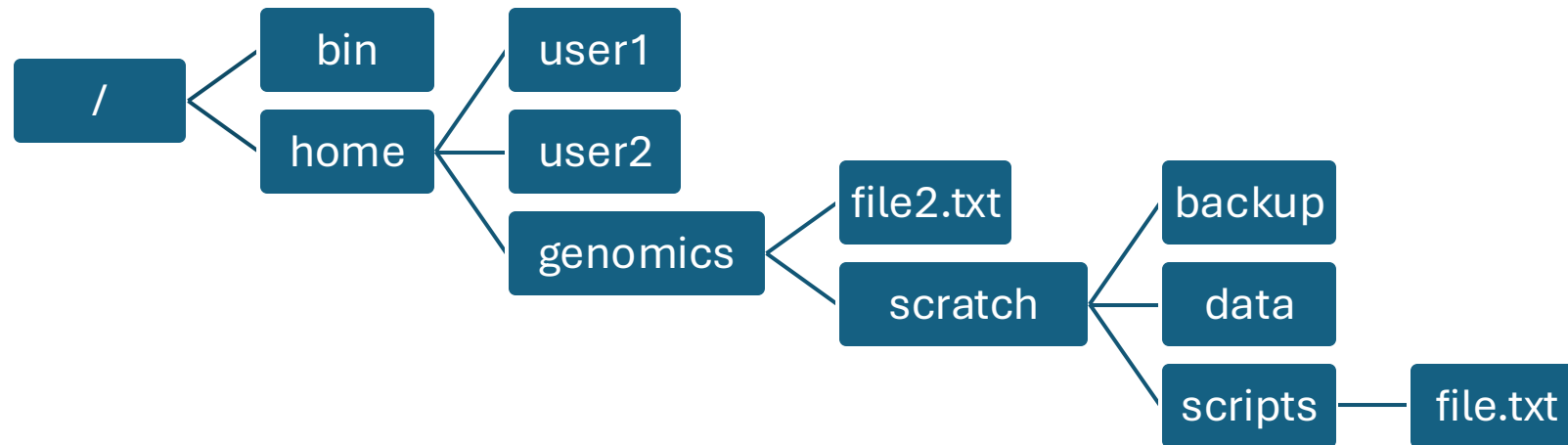
```
kn9 — ssh -X -Y genomics@toko.uncu.edu.ar — 52x9
(base) genomics@toko:~/scratch$ pwd
/home/genomics/scratch
(base) genomics@toko:~/scratch$ ls
backup/  data/  scripts/
(base) genomics@toko:~/scratch$
```



How to navigate the filesystem - paths

You navigate by following the path

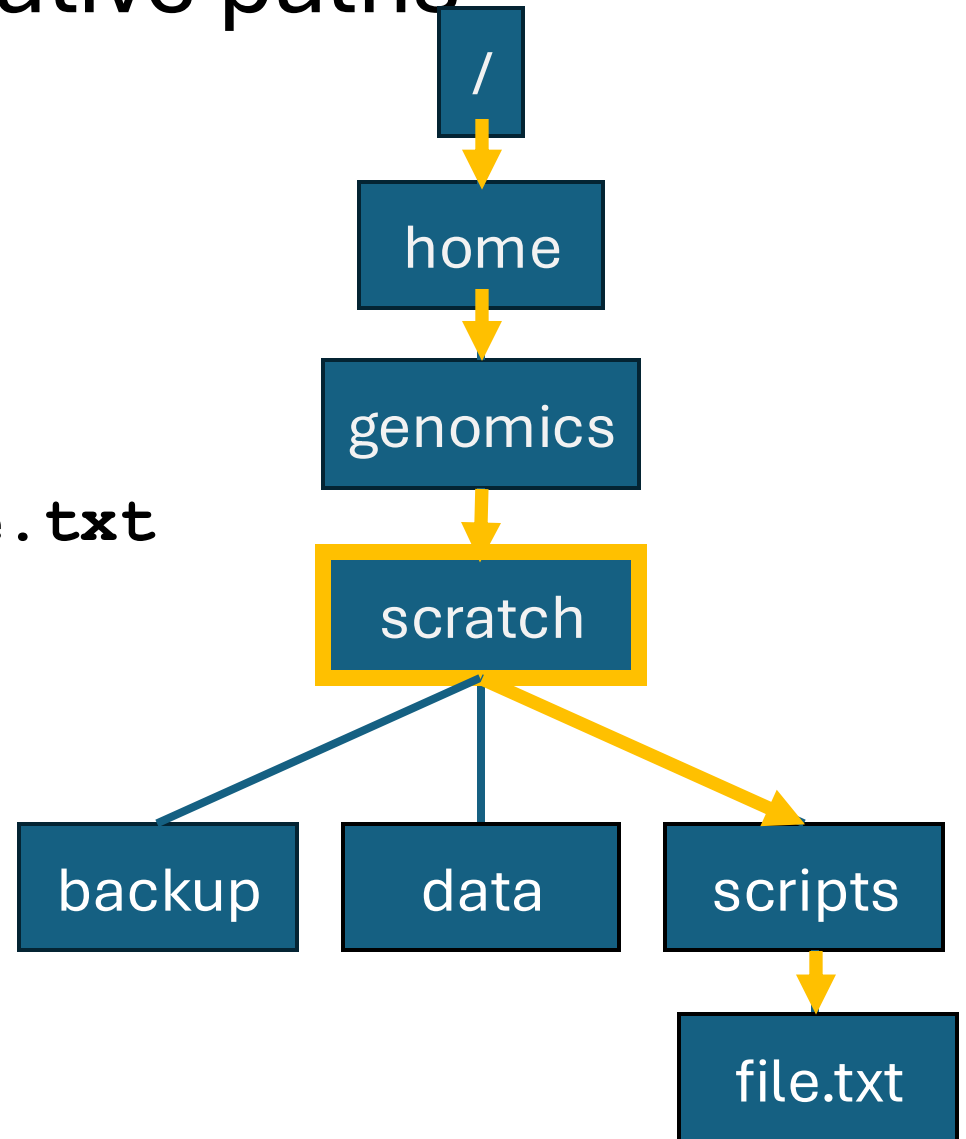
_____ path _____ filename
/home/genomics/scratch/scripts/file.txt



The filesystem - Absolute vs relative paths

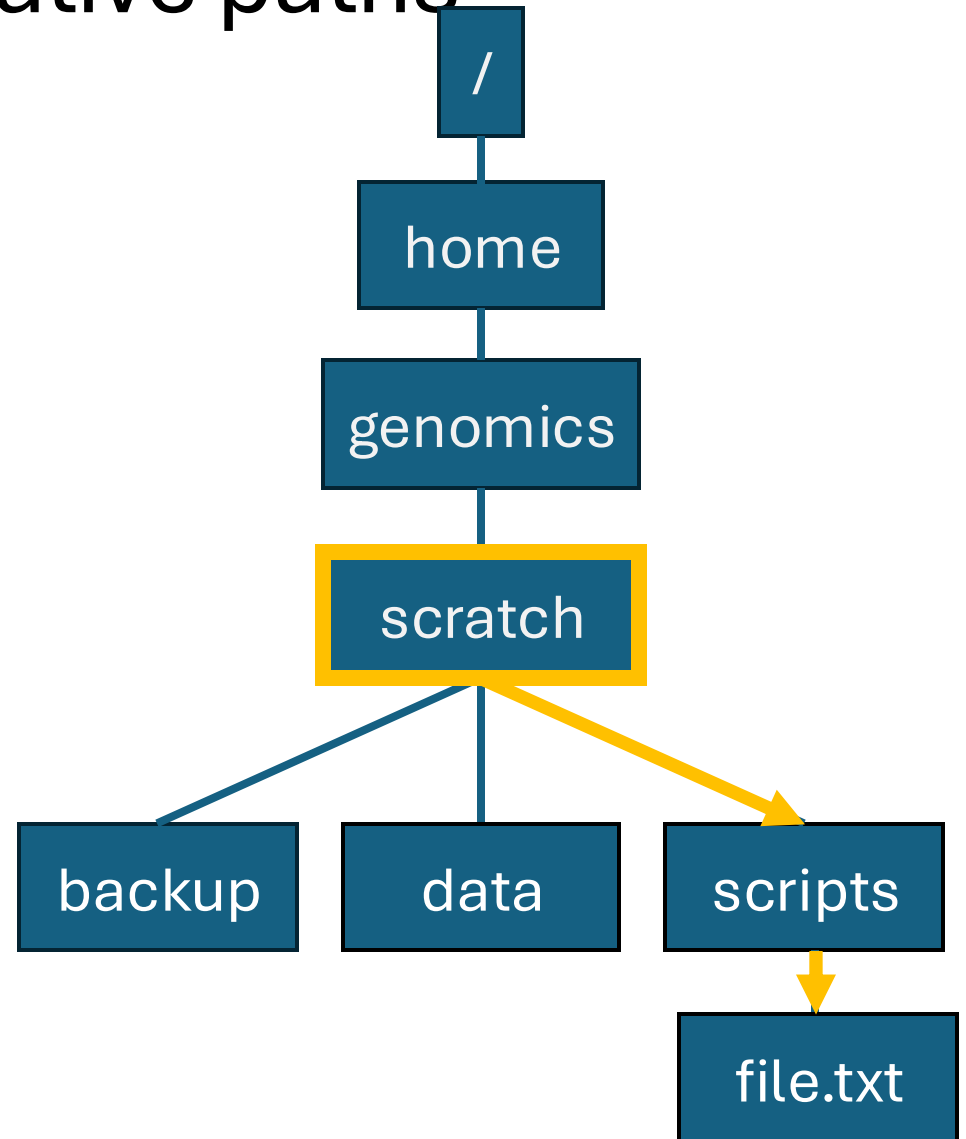
- Absolute path
 - The path from the root (/)

`/home/genomics/scratch/scripts/file.txt`



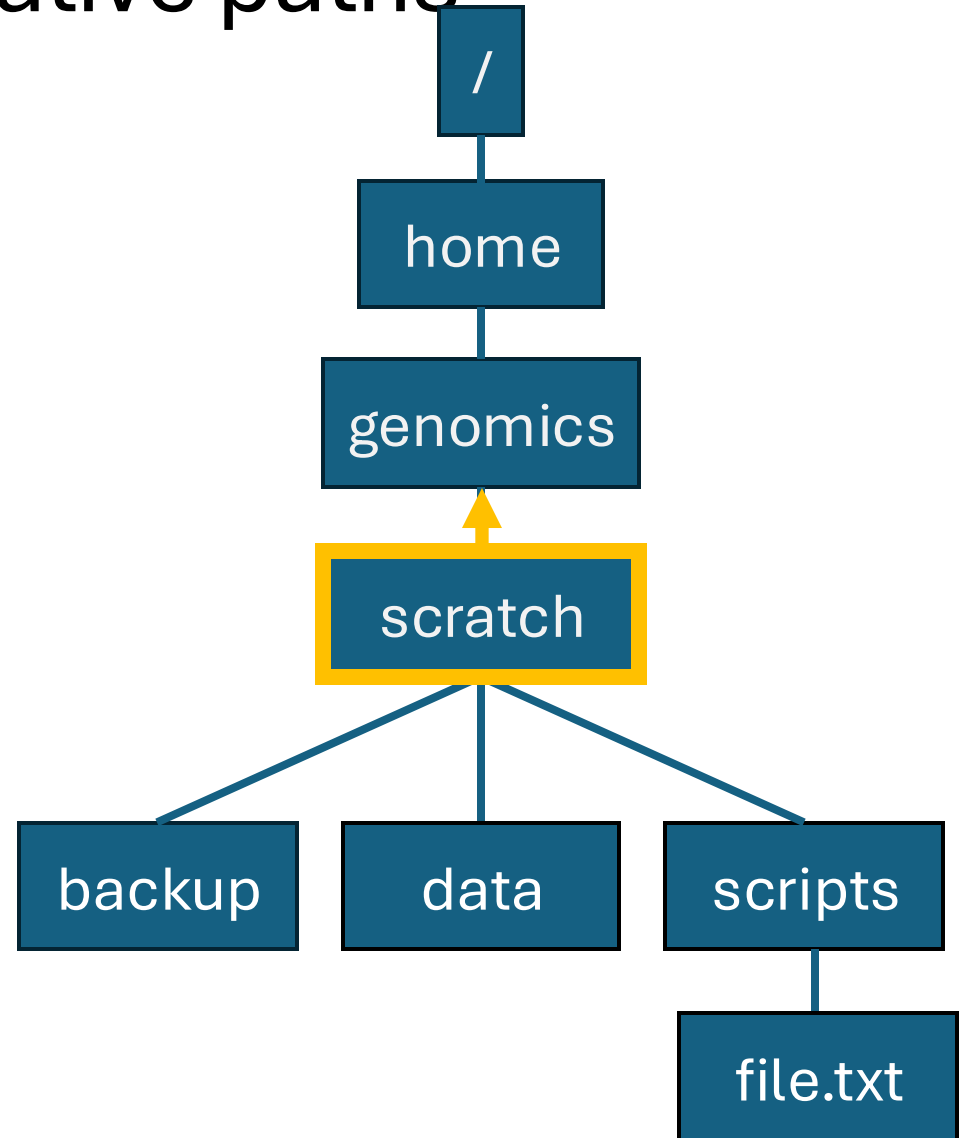
The filesystem - Absolute vs relative paths

- Relative path
 - Current directory
 - Name is shown in the prompt
 - ./ (dot)
 - List content of current directory
 - `ls ./`
 - List content of directory below current directory
 - `ls ./scripts/`



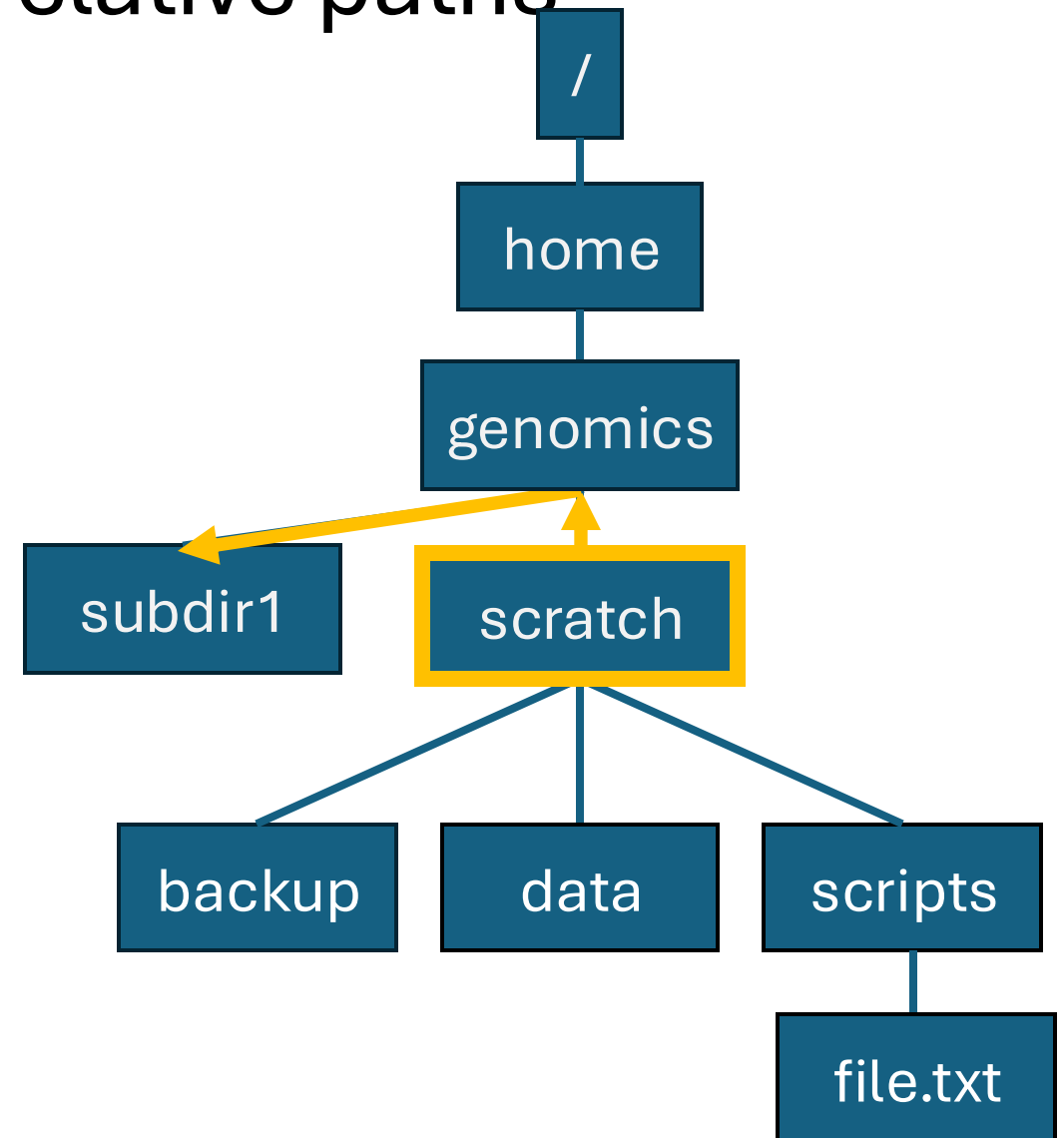
The filesystem - Absolute vs relative paths

- Relative path
 - Parent directory
 - ../ (two dots)
 - List content of parent directory (one level above)
 - `ls ../`



The filesystem - Absolute vs relative paths

- Relative path
 - Parent directory
 - ../ (two dots)
 - List content of parent directory (one level above)
 - `ls ../`
 - List content of another directory
 - `ls ../subdir1/`



Naming conventions for files and directories

- Informative!
- Short
- No spaces or special characters (~~! % # ?~~)
 - Except _ and .
- Files – use correct suffix (.txt, .csv) facilitates usability

Naming conventions

- Informative!
- Short
- No spaces or special characters
(! % # ?)
 - Except _ and .
- Files – use correct suffix (.txt, .csv) facilitates usability
- Keep directories organised
 - Make a new directory for every task
 - Better with multiple directories and fewer files in each directory
 - You will practice this in the course

```
• kn9@farm22-head1:~/comp_gen/04_genome_alignment$ find .
./
./fastga_output
./fastga_output/align_ilMeMeno1_ilMeMeno136
./fastga_output/align_ilMeMeno1_ilMeMeno136/plots
./fastga_output/align_ilMeMeno1_ilMeMeno136/ilMeMeno1_ilMeMeno136.chain.paf
./fastga_output/align_ilMeMeno1_ilMeMeno136/ilMeMeno1_ilMeMeno136.chain.1aln
./fastga_output/align_ilMeMeno1_ilMeMeno136/temp
./fastga_output/align_ilMeMeno1_ilMeMeno136/temp/ilMeMeno136.fa
./fastga_output/align_ilMeMeno1_ilMeMeno136/temp/ilMeMeno136.ilMeMeno136
./fastga_output/align_ilMeMeno1_ilMeMeno136/ilMeMeno1_ilMeMeno136.1aln
./fastga_output/align_ilMeMeno1_ilMeMeno136/intermediate
./fastga_output/align_ilMeMeno1_ilMeMeno136/intermediate/chain_ilMeMeno1ilMeMeno1362000.txt
./fastga_output/align_ilMeMeno1_ilMeMeno136/intermediate/chr_length_ilMeMeno136ilMeMeno12000.txt
./fastga_output/align_ilMeMeno2_ilMeMeno136
./fastga_output/align_ilMeMeno2_ilMeMeno136/ilMeMeno2_ilMeMeno.1aln
./fastga_output/align_ilMeMeno2_ilMeMeno136/plots
./fastga_output/align_ilMeMeno2_ilMeMeno136/plots/synt_ilMeMeno2ilMeMeno13620002025-11-19.pdf
./fastga_output/align_ilMeMeno2_ilMeMeno136/Rplots.pdf
./fastga_output/align_ilMeMeno2_ilMeMeno136/ilMeMeno2_ilMeMeno.chain.1aln
./fastga_output/align_ilMeMeno2_ilMeMeno136/intermediate
./fastga_output/align_ilMeMeno2_ilMeMeno136/intermediate/chain_ilMeMeno2ilMeMeno1362000.txt
./fastga_output/align_ilMeMeno2_ilMeMeno136/intermediate/chr_length_ilMeMeno136ilMeMeno22000.txt
./fastga_output/align_ilMeMeno2_ilMeMeno136/ilMeMeno2_ilMeMeno136.chain.paf
./README
./log
./log/fastga_lsf_output.37727.0.txth
./log/fastga_lsf_output.930028.0.txth
./log/fastga_lsf_error.37727.0.txt
./log/fastga_lsf_error.931237.0.txt
./log/fastga_lsf_error.33037.0.txt
./log/fastga_lsf_output.32886.0.txth
./log/minimap_lsf_error.82993.0.txt
./log/fastga_lsf_output.31591.0.txth
./log/fastga_lsf_output.33037.0.txth
./log/minimap_lsf_error.81109.0.txt
./log/minimap_lsf_output.82556.0.txt
./log/fastga_lsf_error.42839.0.txt
./log/fastga_lsf_error.31591.0.txt
./log/minimap_lsf_output.82993.0.txt
./log/minimap_lsf_output.82604.0.txt
./log/minimap_lsf_error.82556.0.txt
./log/fastga_lsf_output.42839.0.txth
./log/fastga_lsf_error.32886.0.txt
./log/minimap_lsf_output.81109.0.txt
./log/minimap_lsf_error.82604.0.txt
./log/fastga_lsf_error.930028.0.txt
./log/fastga_lsf_output.931237.0.txth
./scripts
./scripts/fastga_specific_genomes_MelMeno1_vs_MelMeno136.bsub
./scripts/fastga_specific_genomes_MelMeno2_vs_MelMeno136.bsub
./scripts/minimap_ilMeMeno1_ilMeMeno136.bsub
./minimap_output
./minimap_output/minimap_ilMeMeno1_ilMeMeno136
./minimap_output/minimap_ilMeMeno1_ilMeMeno136/plots
./minimap_output/minimap_ilMeMeno1_ilMeMeno136/plots/synt_ilMeMeno1ilMeMeno1362025-11-19.pdf
./minimap_output/minimap_ilMeMeno1_ilMeMeno136/ilMeMeno1_ilMeMeno136.paf
./minimap_output/minimap_ilMeMeno1_ilMeMeno136/Rplots.pdf
./minimap_output/minimap_ilMeMeno1_ilMeMeno136/ilMeMeno1_ilMeMeno136.err
./minimap_output/minimap_ilMeMeno1_ilMeMeno136/intermediate
./minimap_output/minimap_ilMeMeno1_ilMeMeno136/intermediate/chr_length_ilMeMeno136ilMeMeno1.txt
./minimap_output/minimap_ilMeMeno1_ilMeMeno136/intermediate/chain_ilMeMeno1ilMeMeno136.txt
./minimap_output/minimap_ilMeMeno2_ilMeMeno136
./minimap_output/minimap_ilMeMeno2_ilMeMeno136/plots
./minimap_output/minimap_ilMeMeno2_ilMeMeno136/plots/synt_ilMeMeno2ilMeMeno1362025-11-19.pdf
./minimap_output/minimap_ilMeMeno2_ilMeMeno136/ilMeMeno2_ilMeMeno136.err
./minimap_output/minimap_ilMeMeno2_ilMeMeno136/ilMeMeno2_ilMeMeno136.paf
./minimap_output/minimap_ilMeMeno2_ilMeMeno136/Rplots.pdf
./minimap_output/minimap_ilMeMeno2_ilMeMeno136/intermediate
./minimap_output/minimap_ilMeMeno2_ilMeMeno136/intermediate/chr_length_ilMeMeno136ilMeMeno2_mod.txt
./minimap_output/minimap_ilMeMeno2_ilMeMeno136/intermediate/chain_ilMeMeno2ilMeMeno136.txt
./minimap_output/minimap_ilMeMeno2_ilMeMeno136/intermediate/chr_length_ilMeMeno136ilMeMeno2.txt
• kn9@farm22-head1:~/comp_gen/04_genome_alignment$
```

17 directories, 53 files

Running commands - structure

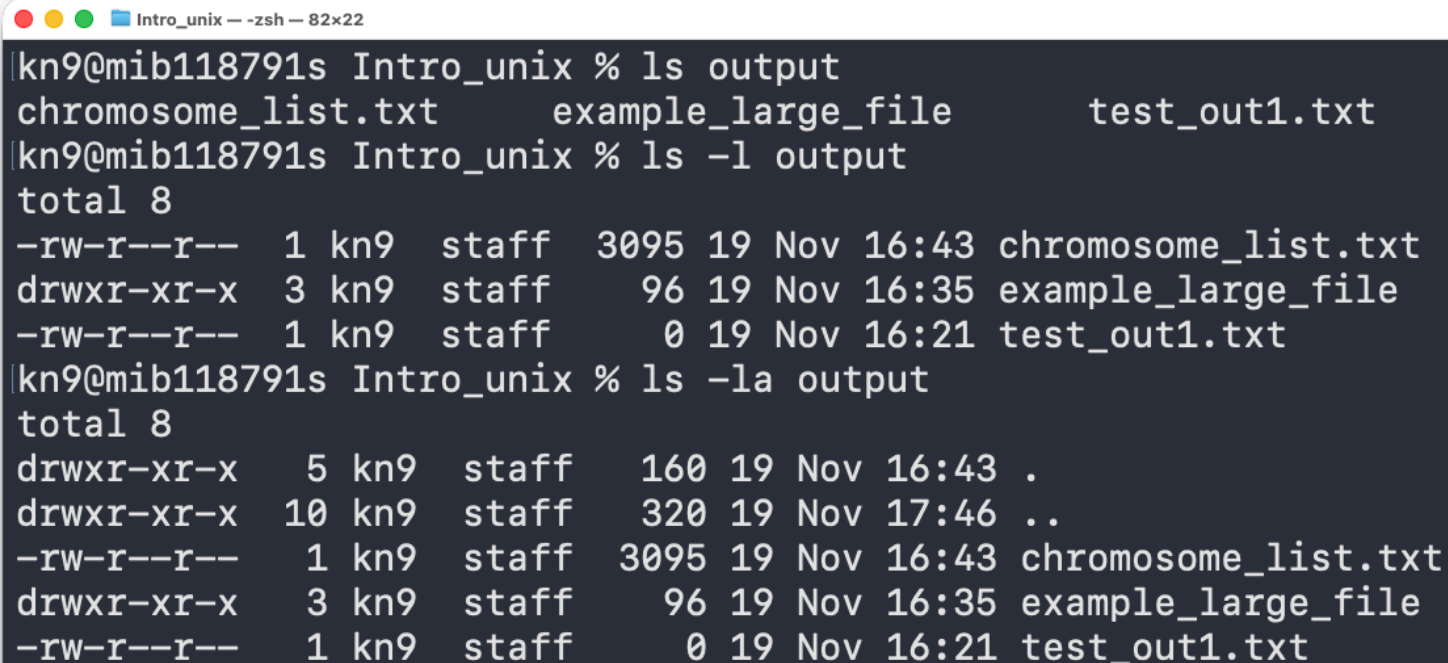
command [options] <arguments>

- Command is the program you want to use
- Options are flags that specifies what you want the program to do
 - usually with a hyphen(dash) – or two --
 - -h , -f
 - --help, --file
- Arguments specifies what you want the program to perform the task with
 - example filename or path

Running commands - structure

- command [options] <arguments>

```
ls -l output
```



A terminal window titled "Intro_unix -- zsh -- 82x22" showing the execution of two commands. The first command is `ls output`, which lists three files: `chromosome_list.txt`, `example_large_file`, and `test_out1.txt`. The second command is `ls -l output`, which provides a detailed listing of the same three files, including permissions, owner, group, size, and modification date. The third command is `ls -la output`, which lists all files and directories, including the current directory (`.`) and the parent directory (`..`).

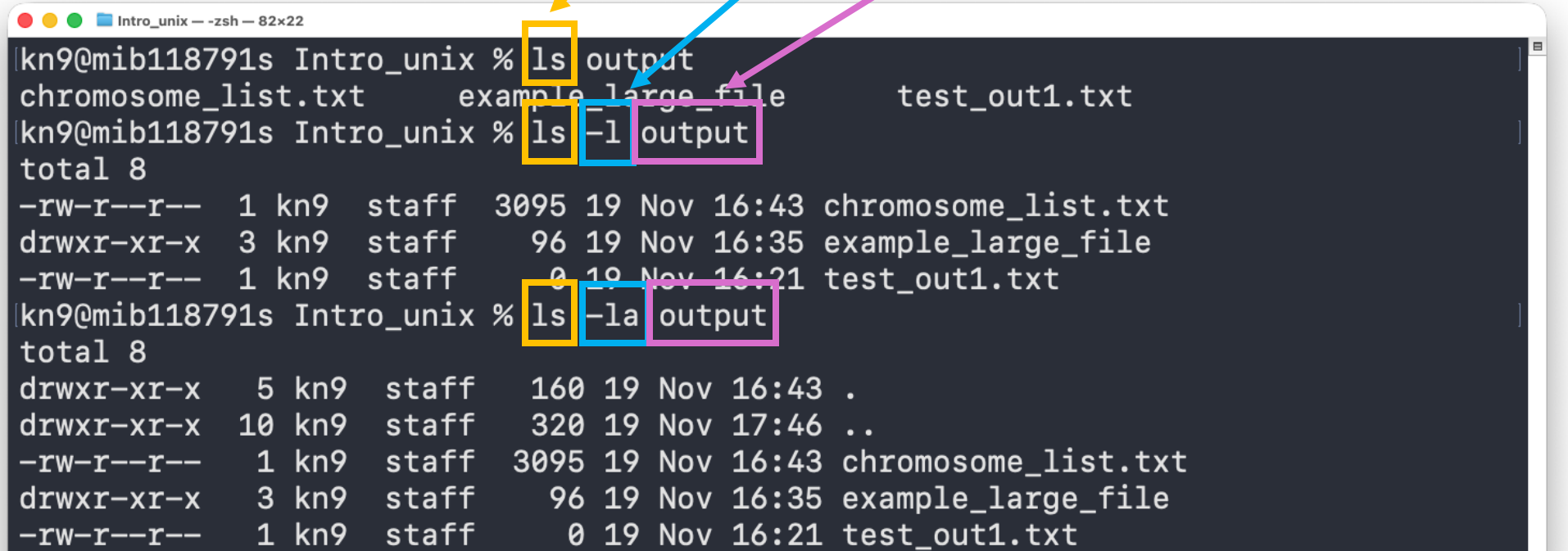
```
kn9@mib118791s Intro_unix % ls output
chromosome_list.txt      example_large_file      test_out1.txt
kn9@mib118791s Intro_unix % ls -l output
total 8
-rw-r--r--  1 kn9  staff  3095 19 Nov 16:43 chromosome_list.txt
drwxr-xr-x  3 kn9  staff   96 19 Nov 16:35 example_large_file
-rw-r--r--  1 kn9  staff    0 19 Nov 16:21 test_out1.txt
kn9@mib118791s Intro_unix % ls -la output
total 8
drwxr-xr-x  5 kn9  staff  160 19 Nov 16:43 .
drwxr-xr-x 10 kn9  staff  320 19 Nov 17:46 ..
-rw-r--r--  1 kn9  staff  3095 19 Nov 16:43 chromosome_list.txt
drwxr-xr-x  3 kn9  staff   96 19 Nov 16:35 example_large_file
-rw-r--r--  1 kn9  staff    0 19 Nov 16:21 test_out1.txt
```


Running commands - structure

- command [options] <arguments>

```
ls -l output
```

command options <arguments>



A terminal window titled "Intro_unix -- zsh -- 82x22" showing the execution of the `ls` command. The prompt is `kn9@mib118791s Intro_unix %`. The first command is `ls output`, which lists three files: `chromosome_list.txt`, `example_large_file`, and `test_out1.txt`. The second command is `ls -l output`, which shows detailed permissions, owner, group, size, and date for the same files. The third command is `ls -la output`, which shows all files, including `.` and `..`. Annotations with colored boxes and arrows identify the components of the commands: a yellow box around `ls` is labeled "command", a blue box around `-l` is labeled "options", and a pink box around `output` is labeled "<arguments>".

```
kn9@mib118791s Intro_unix % ls output
chromosome_list.txt  example_large_file  test_out1.txt
kn9@mib118791s Intro_unix % ls -l output
total 8
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kn9@mib118791s Intro_unix % ls -la output
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drwxr-xr-x  3 kn9  staff   96 19 Nov 16:35 example_large_file
-rw-r--r--  1 kn9  staff    0 19 Nov 16:21 test_out1.txt
```

Running commands - getting help

- Manual
 - `man <command>`
 - exit manual: `q`
- Type the command followed by
 - `--help` (or `-h`)
- Type the command without options or arguments (not always)
- Cheat Sheet
- Google ([Stackoverflow](#))
- AI (beware of errors!)

Running commands - getting help

- Manual
 - `man <command>`
 - exit manual: `q`
 - Type the command followed by
 - `--help` (or `-h`)
 - Type the command without options or arguments (not always)
 - Cheat Sheet
 - Google ([Stackoverflow](#))
 - AI (beware of errors!)
- `man ls`
 - To exit type `q`

Version control and backup

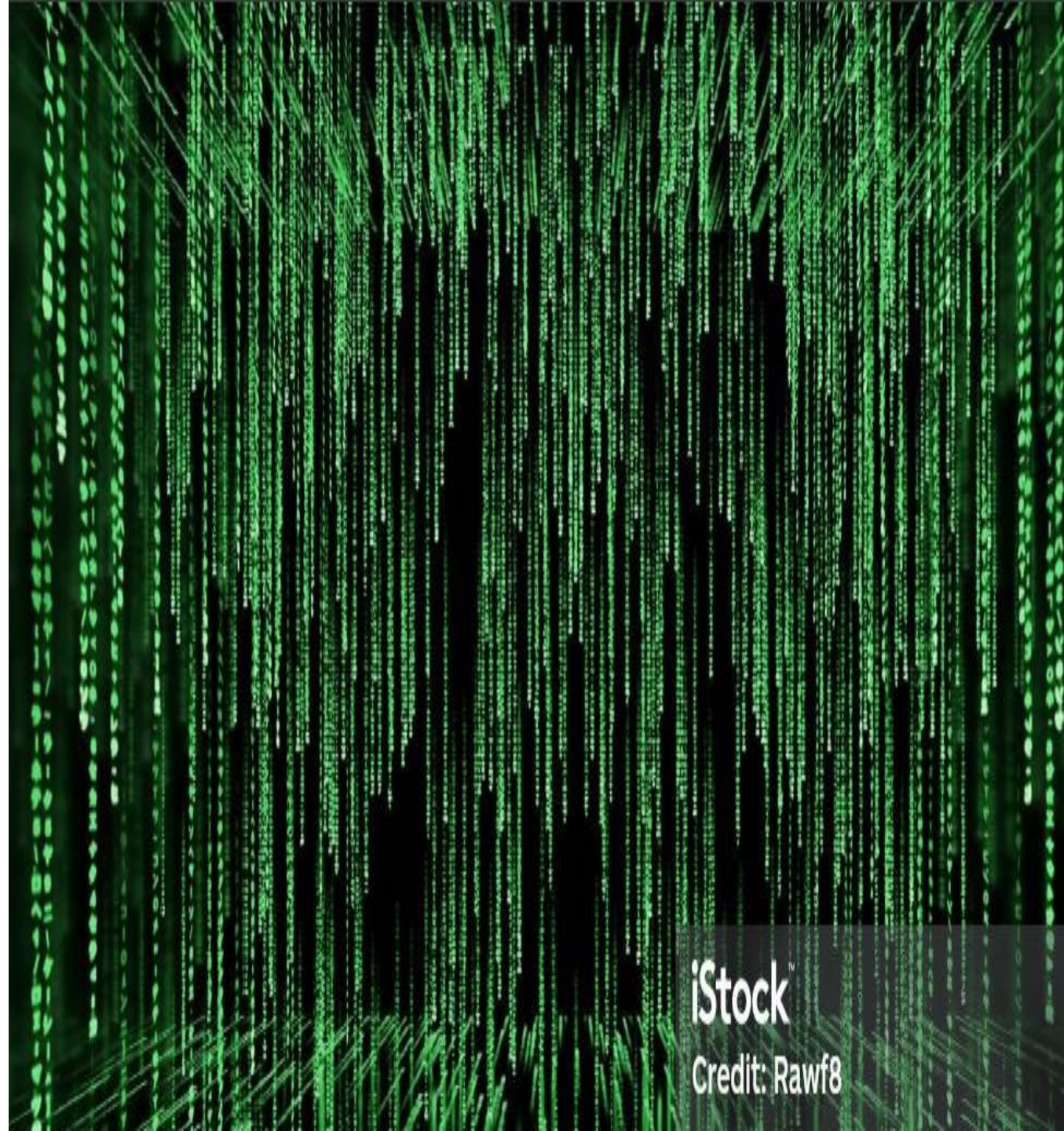


- There is no undo button!
 - Be careful when running commands to delete, copy and change file content
- Version control
 - Git
 - Stores the history of files, allows you to go back to earlier versions
- Backup
 - Always backup
 - Raw data (long term storage, external hard drives, public repositories)
 - Scripts (GitHub or other repositories)
 - Good to backup
 - Important (costly) intermediate files
 - Result files



Running commands

- Stop process
 - Ctrl + C



Good Practices

- Use meaningful file names
- Keep directories organized
 - Make a new directory for every task
 - Better with more directories – less files in each directory
- Simple is powerful
 - The power of simple shell commands in combinations can do very complex tasks
- Collect your commands in executable and reusable scripts
- Open source and code sharing
- Many ways to do what you want – ask google/AI!!
- Backup!



Practice

Basic

- Navigating directories

cd, ls, cat, cp, mv

More

- Working with files

wc, grep, sed, pipe

Advanced

awk, variables and loops